

Brian Guadalupe Tellez Escobar

Computer Systems Engineering Student · Full-Stack Developer & DevOps

Tijuana, Baja California, Mexico | Available for remote and on-site work

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LinkedIn

PROFILE

Ninth-semester Computer Systems Engineering student (GPA **93.64/100**) with a portfolio of more than ten personal projects running in production, ranging from full-stack web applications to AI systems and self-hosted infrastructure.

What sets me apart is command of the full cycle: I design and assemble the server hardware, administer the Linux system, containerize the application, expose it securely, and write both the backend and the frontend. I learn independently and take projects all the way to working with real users — not just to the prototype stage. I am seeking a **professional residency** (internship) or a **junior** position where I can contribute this end-to-end execution capability.

FEATURED PROJECTS

NoteCore — Academic Organisation Platform [\[GitHub\]](#)

Personal project · August 2026 – Present · In production · notecore.ourocore.net

Android and web app bringing class schedule, absence tracking and assignment planning together, with the schedule as the backbone. In use by real students and ready for **Google Play**.

- Three-layer monorepo in strict TypeScript: Android app in React Native with Expo, web in Next.js 16 and API in Fastify, sharing types and logic.
- All business logic lives in the API and is never duplicated in the clients, so app and web cannot drift apart.
- Absence tracking with a suggested limit derived from the 80% attendance rule, and schedule sharing by QR, code or link.
- A family of seven native home-screen widgets in Kotlin that refresh on their own without opening the app.
- **Results:** eight real accounts in production from day one; signed .aab and store listing ready for Google Play.

React Native · Expo · Next.js 16 · Fastify · PostgreSQL · Drizzle ORM · Kotlin · TypeScript · Docker

ADA — AI Voice Assistant for University Campus [\[GitHub\]](#)

Instituto Tecnológico de Tijuana · February 2026 – Present · 2-person team

A standalone physical kiosk that answers student questions through natural speech in under 8 seconds, built on a two-node **Fog Computing** architecture.

- Complete AI pipeline: Whisper transcription → RAG semantic retrieval (ChromaDB) → Llama 3 comprehension → Mexican Spanish speech synthesis.
- Edge node on Raspberry Pi 5 with a Flutter interface, physical *push-to-talk* button via GPIO, and word-by-word synchronized subtitles.
- Fog node on a Ryzen server running containerized microservices communicating over WebSockets.
- **Results:** ~95% transcription accuracy (WER) under moderate noise; validated with real students.

Flutter · Python FastAPI · faster-whisper · Ollama/Llama 3 · ChromaDB · PiperTTS · Raspberry Pi 5 · WebSockets

Koko — P2P Co-presence Application [\[koko.ourocore.net\]](https://koko.ourocore.net)

Personal project · Version 0.9.0, in active use

A desktop application that lets two people at a distance share presence: each controls a transparent avatar that lives on the other person's desktop.

- Transparent, *always-on-top* Electron window with *click-through*, so it never interrupts the user's work.
- Direct **WebRTC P2P connection** — content never touches a server and is never stored.
- Cross-platform distribution (Windows NSIS without UAC, Linux AppImage) with auto-update via a self-hosted feed.
- Solved real platform constraints: dropped WebGL due to incompatibility with transparent windows on Windows; forced *HTTP polling* to prevent Cloudflare from dropping events.

Electron 39 · React 19 · TypeScript · WebRTC · Socket.io · Node.js · Cloudflare Tunnel

Wander — Gamer Profile Platform

Personal project · In active development

A social platform where players build their gaming identity in a single link, with data imported automatically from platforms such as Steam.

- Profile editor built from reorderable blocks with full theme control.
- Authentication with **argon2id** and JWT sessions in *httpOnly* cookies.
- Privacy by design: granular per-integration permissions and genuine deletion on unlinking.

React 19 · Vite · TypeScript · Tailwind 4 · Express 5 · Prisma 7 · PostgreSQL 17 · Docker

ZAGA Distribuciones — Corporate Website (real client) [\[zagadistribuciones.com\]](https://zagadistribuciones.com)

Freelance

A bilingual website for an electrical supply distributor in Tampico, Tamaulipas, Mexico.

- Static site generation with Astro, deployed on Cloudflare Pages.
- Spanish/English internationalization with a language-annotated sitemap.
- Real optimization: fonts self-hosted at build time, zero external requests in production.

Astro 7 · Tailwind CSS 4 · Cloudflare Pages · Wrangler

Kanua Petto — Cross-platform Virtual Pet [\[GitHub\]](https://github.com)

Personal project · September – December 2025

A virtual pet whose state persists in the cloud and syncs instantly between PC and Android.

- Native client in Godot 4.5 with C# (.NET), compiled for Windows and Android.
- REST API with JWT authentication, bcrypt-hashed passwords, and MongoDB persistence.
- Affinity system: the pet's personality evolves according to how it is treated.

Godot 4.5 · C# · Node.js/Express · MongoDB · Docker Compose · Cloudflare Tunnel

Homelab Infrastructure

Personal project · August 2025 – Present

A bare-metal server I designed, assembled and administer myself, hosting all of my projects in production.

- Hardware: AMD Ryzen 5 9600X · 64 GB DDR5 · 1 TB NVMe SSD + 4 TB HDD · 4U rack.
- Ubuntu Server administration via command line, remote access over SSH.
- Orchestration of Docker containers, databases, and game servers for communities.
- Secure service exposure through **Cloudflare Tunnels (Zero Trust)**, with no public IP.

OTHER PROJECTS

IRIS	Web platform with media manager, digital economy and real-time multiplayer games	React · FastAPI · MongoDB · WebSockets
SGV	Multi-tenant SaaS for residential property management, with financial control and audit logging	React · FastAPI · PostgreSQL · Docker
Shokan	Push-notification PWA for rallying gaming groups (Web Push/VAPID, no Firebase)	React · Express · Prisma · PostgreSQL
Discord Bots	Audio ecosystem with an OAuth2 web panel and EBU R128 loudness normalization	Python · FastAPI · discord.py · FFmpeg
Portfolio	Static site with custom i18n built without libraries, dynamic themes and SSG	Next.js 16 · Tailwind 4 · Docker · Nginx

EDUCATION

B.Eng. in Computer Systems Engineering
Instituto Tecnológico de Tijuana (ITT) · Currently in 9th semester · Cumulative GPA: **93.64/100**
Professional residency (internship) scheduled for the current semester.

Technical Track in Architectural Drafting
Preparatoria Federal Lázaro Cárdenas (High School) · Graduated 2021 · GPA: 86/100

TECHNICAL SKILLS

Languages	Python · JavaScript / TypeScript · C# (.NET) · Kotlin · SQL · Dart · HTML / CSS · Haskell · Prolog
Frontend	React 19 · Next.js · Astro · Vite · Tailwind CSS · Flutter · Electron · Zustand
Backend	FastAPI · Node.js / Express · Prisma · SQLAlchemy · REST APIs · WebSockets · WebRTC · JWT
Databases	PostgreSQL · MongoDB · Firebase Realtime Database · ChromaDB
Infrastructure	Docker & Docker Compose · Linux (Ubuntu Server) · Nginx · Cloudflare Tunnels (Zero Trust) · Git & GitHub · systemd · Networking (SSH, Samba, FTP)
AI & Data	RAG · Embeddings · Whisper (ASR) · TTS · LLMs (Ollama, Llama 3) · Data evaluation and annotation
Hardware & IoT	Bare-metal server assembly · Raspberry Pi · Arduino / ESP32 · Fog Computing · 3D modeling and printing (Blender)

LANGUAGES

Spanish	Native — spoken and written
English	B1 — advanced reading and listening comprehension in technical contexts; spoken production actively developing

PERSONAL STRENGTHS

Self-directed learning · Deep work · Creative problem solving · Critical and analytical thinking · Adaptability · Teamwork · End-to-end ownership of personal projects